riffle

Reflow content into new margins in the middle of a page

Manual · version 0.2.1

1 Introduction

This package reflows a stream of content into new page margins, starting exactly where the content appears and **without forcing a page break**. It is designed for documents that mix an asymmetric (book-style, wide outer margin) layout with blocks that read better at a different width.

One function covers every direction:

```
#margin-reflow(
  content,
  columns: (count: 1, gutter: 4%),
  margin: auto,
  footnote-style: none,
  set-page-margin: true,
)
```

- `columns` chooses the number of columns (and the gutter) of the reflowed content,
- `margin` chooses the margins to switch to,
- `footnote-style` styles the footnote entries, which the reflow lays out itself,
- `set-page-margin` chooses between the fast path (`true`, using `set page(...)`) and the fully hand-split path (`false`).

The reflow pulls footnotes out of the flow and renders them at the bottom of the reflowed block, preserves footnote numbering and `ref`s, lays the reflowed block flush to the bottom of the available height, and keeps the new margins on the pages that follow. It is a `context` function, so it inspects the current page geometry and position itself.

2 Installation

The package is available on Typst Universe:

```
#import "@preview/riffle:0.2.1": margin-reflow
```

3 Requirements

Use the function in normal document flow, on pages whose size and margins are known: the page width and height must be concrete lengths.

The first-line indentation of the reflowed content follows the ambient `par.first-line-indent`, so set it explicitly:

```
#set par(first-line-indent: (amount: 2em, all: true), justify: true)
```

4 API reference

4.1 `margin-reflow`

Reflows `content` into one or more columns with the margins given by `margin`, starting at the position where the function is called and without starting a new page before the content.

```
#margin-reflow(
  content,
  columns: (count: 1, gutter: 4%),
  margin: auto,
  footnote-style: none,
  set-page-margin: true,
)
```

- `content` **(content)**: the content to reflow. Paragraph breaks are honored, so a paragraph never starts in one column and continues in the next.
- `columns` **(dictionary, default `(count: 1, gutter: 4%)`)**: forwarded to the column layout. Only `count` **(int, default `1`)** and `gutter` **(length, default `4%` of the reflow width)** are meaningful.
- `margin` **(auto, length, dictionary or `"symmetric"`, default `auto`)**: the horizontal margin to switch to.
  - a page-margin dictionary such as `(inside: 1cm, outside: 3cm)` or `(left: 1cm, right: 3cm)`,
  - a plain length, which yields equal left/right margins,
  - `"symmetric"`, which uses `min(inside, outside)` of the current page on both sides,
  - `auto`, which inherits the current page margin.

  Only the horizontal margins are taken from this value; the top and bottom margins of the current page are preserved.
- `footnote-style` **(none, dictionary or function, default `none`)**: styling for the manually laid-out footnote entries.
  - `none`: the inherited defaults,
  - a dictionary: applied as `set text(...)` to the body of each entry, so it takes `text`'s arguments such as `size`, `fill` or `style`,
  - a function: applied to the body of each entry directly.

  The `gap`, `indent`, `clearance` and `separator` of the entries are taken from the ambient `footnote.entry` settings, so configure those with `#set footnote.entry(...)` as usual.
- `set-page-margin` **(bool, default `true`)**: how the new margins are applied.
  - `true`: the margins of the pages **after** the reflow are switched with `set page(...)`; only the current page is split by hand. Fast, and the recommended default. Works with any `count`.
  - `false`: no `set page(...)` is used at all, so every page is split by hand and the margin change is emulated with padding. Slower, and it writes one text block per page, which is useful when the document's page settings must not change. Only `count: 1` (`columns` without `count`, or `count: 1`) is allowed: with more than one column the hand-split path cannot change the layout within a page, so the last page would still be filled as a single column. The combination is rejected with an assertion.

5 Usage notes

5.1 Footnotes

Footnotes are removed from the reflowed content and rendered by the package, so their entries line up with the reflowed block instead of the page width. Numbering and `ref`s keep working. Use `footnote-style` to style the body of the entries, and `#set footnote.entry(...)` for their spacing (`gap`, `indent`, `clearance`, `separator`), which the package reads from the ambient settings.

5.2 First-line indentation

Each column chunk is placed in its own box, so the ambient `par.first-line-indent` is applied explicitly by the package: set `first-line-indent: (amount: ..., all: true)` if the first paragraph of a chunk should be indented too.

5.3 Page numbers and counters

The counters Typst resolves at layout time, such as `counter(page)` and custom counters, are unaffected: they follow document order across the reflow.

6 Examples

Each example below is a complete, compilable file. Its two rendered pages are shown next to it. In the page images, the **blue** dashed line marks the symmetric content boundary and the **orange** dashed line the asymmetric one.

7 Two columns on an asymmetric page

```
// Example: two columns across the symmetric width on an asymmetric page.
//
// Compile to page images with:
//   typst compile -f png --ppi 144 examples/column-flow.typ "doc/images/column-flow-
{p}.png"
//
// The prose switches to Chinese when compiled with `--input lang=zh`, which is
// how `doc/manual-zh.typ` renders its figures.

#import "../lib.typ": margin-reflow
// Removes the spaces that source line breaks would otherwise insert between
// CJK characters.
#import "@preview/cjk-unbreak:0.2.3": remove-cjk-break-space
#import "assets/guides.typ": guides

#let lang = sys.inputs.at("lang", default: "en")

#set page(
  paper: "a5",
  margin: (inside: 1cm, outside: 2.5cm),
  numbering: "1",
  background: guides(),
)
#set text(font: ("New Computer Modern", "SimSun"), size: 9pt, lang: lang)
#set par(first-line-indent: (amount: 2em, all: true), justify: true, leading: 0.6em)

// Body text at the normal asymmetric margins.
#let preamble = if lang == "zh" { [中文与英文混排的正文示例。] } else { [#lorem(45)] }

#preamble

#v(1em)

#let reflow-body = if lang == "zh" { remove-cjk-break-space[

    重排从这里开始，此时页面仍处于非对称页边距之下。`margin-reflow` 会把随后的内容
    排入横跨对称宽度的两栏，并让其后的页面沿用该页边距。
    #footnote[位于重排内容开头的脚注。]

    #lorem(230)

    中文与英文混排 testing mixed CJK and Latin reflow across columns。
    假文假文假文假文假文假文假文假文假文假文假文假文假文假文假文假文。

    #lorem(250)
] } else {
```

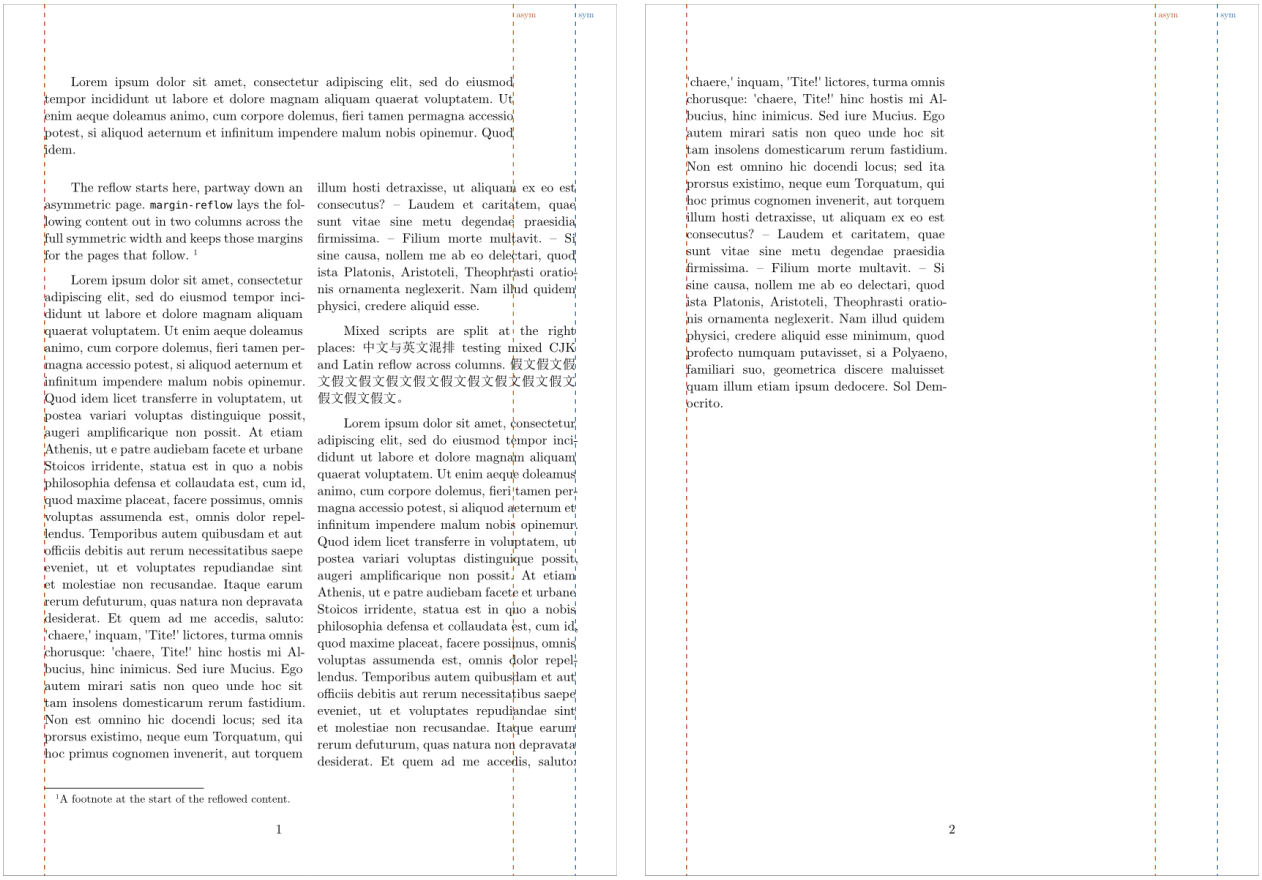
```
  [
    The reflow starts here, partway down an asymmetric page. `margin-reflow` lays
    the following content out in two columns across the full symmetric width and
    keeps those margins for the pages that follow.
    #footnote[A footnote at the start of the reflowed content.]

    #lorem(230)

    Mixed scripts are split at the right places: 中文与英文混排 testing mixed CJK
    and Latin reflow across columns.
    假文假文假文假文假文假文假文假文假文假文假文假文假文假文假文假文。

    #lorem(250)
  ]
}

#margin-reflow(
  columns: (count: 2, gutter: 10pt),
  margin: "symmetric",
)[#reflow-body]
```



Rendered output — page 1 (left) and page 2 (right). The reflowed content spans two pages, so the behaviour on the following pages is visible.

8 One column on an asymmetric page

```
// Example: one column across the symmetric width on an asymmetric page.
//
```

```
// Compile to page images with:
//   typst compile -f png --ppi 144 examples/single-flow.typ "doc/images/single-flow-
{p}.png"
//
// The prose switches to Chinese when compiled with `--input lang=zh`, which is
// how `doc/manual-zh.typ` renders its figures.

#import "../lib.typ": margin-reflow
// Removes the spaces that source line breaks would otherwise insert between
// CJK characters.
#import "@preview/cjk-unbreak:0.2.3": remove-cjk-break-space
#import "assets/guides.typ": guides

#let lang = sys.inputs.at("lang", default: "en")

#set page(
  paper: "a5",
  margin: (inside: 1cm, outside: 2.5cm),
  numbering: "1",
  background: guides(),
)
#set text(font: ("New Computer Modern", "SimSun"), size: 9pt, lang: lang)
#set par(first-line-indent: (amount: 2em, all: true), justify: true, leading: 0.6em)

// Body text at the normal asymmetric margins.
#let preamble = if lang == "zh" { [中文与英文混排的正文示例。] } else { [#lorem(45)] }

#preamble

#v(1em)

#let reflow-body = if lang == "zh" { remove-cjk-break-space[

    把 `count` 设为 1 时，重排后的内容占据一整栏，横跨整个对称内容宽度，而不再分成两栏。
    #footnote[位于重排内容开头的脚注。]

    #lorem(240)

    中文与英文混排 testing mixed CJK and Latin reflow。
    假文假文假文假文假文假文假文假文假文假文假文假文假文假文假文假文假文。

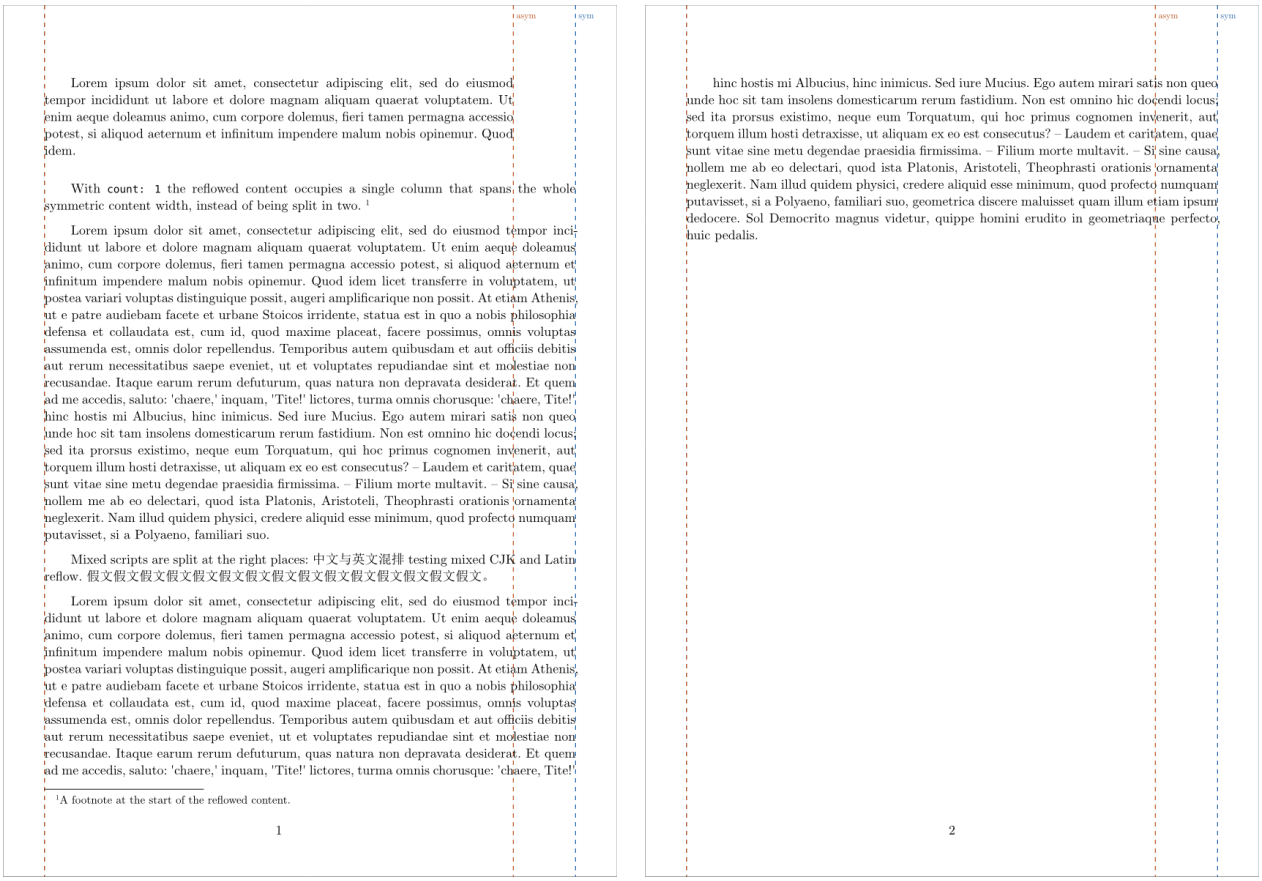
    #lorem(260)
] } else {
  [
    With `count: 1` the reflowed content occupies a single column that spans the
    whole symmetric content width, instead of being split in two.
    #footnote[A footnote at the start of the reflowed content.]

    #lorem(240)

    Mixed scripts are split at the right places: 中文与英文混排 testing mixed CJK
    and Latin reflow.
    假文假文假文假文假文假文假文假文假文假文假文假文假文假文假文假文假文。

    #lorem(260)
  ]
}
```

```
#margin-reflow(
  columns: (count: 1),
  margin: "symmetric",
)[#reflow-body]
```



Rendered output — page 1 (left) and page 2 (right). A single column across the symmetric width.

9 Returning to an asymmetric layout

```
// Example: switch from symmetric margins to an asymmetric layout.
//
// Compile to page images with:
//   typst compile -f png --ppi 144 examples/asymmetric-flow.typ "doc/images/asymmetric-flow-{p}.png"
//
// The prose switches to Chinese when compiled with `--input lang=zh`, which is
// how `doc/manual-zh.typ` renders its figures.

#import "../lib.typ": margin-reflow
// Removes the spaces that source line breaks would otherwise insert between
// CJK characters.
#import "@preview/cjk-unbreak:0.2.3": remove-cjk-break-space
#import "assets/guides.typ": guides

#let lang = sys.inputs.at("lang", default: "en")

#set page(
  paper: "a5",
```

```
  margin: 1.5cm,
  numbering: "1",
  background: guides(sym: (1.5cm, 1.5cm), asym: (1cm, 2.5cm)),
)
#set text(font: ("New Computer Modern", "SimSun"), size: 9pt, lang: lang)
#set par(first-line-indent: (amount: 2em, all: true), justify: true, leading: 0.6em)

// Body text at symmetric margins.
#let preamble = if lang == "zh" { [对称页边距下的正文示例。] } else { [#lorem(40)] }

#preamble

#v(1em)

#let reflow-body = if lang == "zh" { remove-cjk-break-space[

    页面开始时是对称页边距；从这里起内容收窄，改用传给 `margin` 的非对称页边距
    （内侧 1cm、外侧 2.5cm），其后各页沿用该版式。
    #footnote[位于重排内容开头的脚注。]

    #lorem(240)

    中文与英文混排 testing mixed CJK and Latin reflow。
    假文假文假文假文假文假文假文假文假文假文假文假文假文假文假文假文。

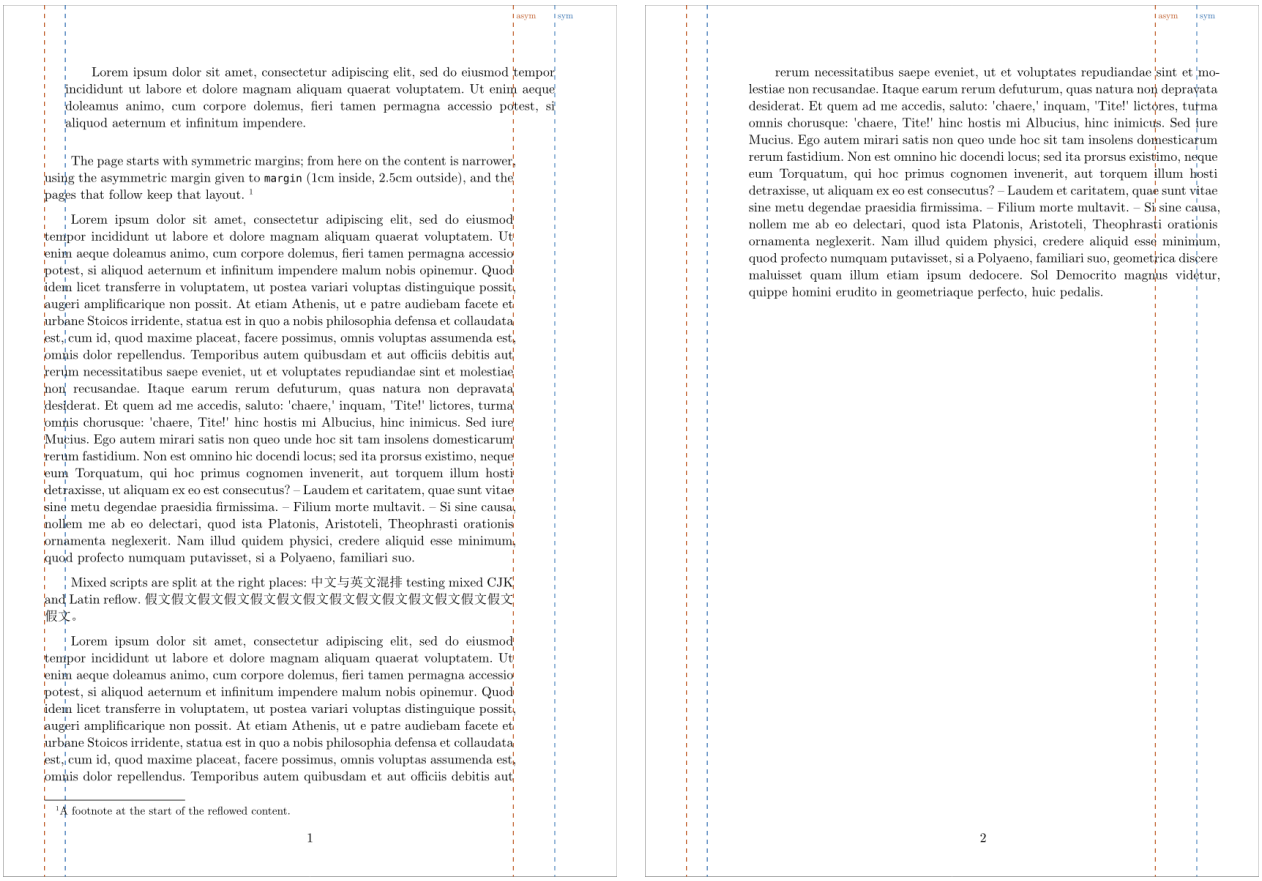
    #lorem(260)
] } else {
  [
    The page starts with symmetric margins; from here on the content is narrower,
    using the asymmetric margin given to `margin` (1cm inside, 2.5cm outside),
    and the pages that follow keep that layout.
    #footnote[A footnote at the start of the reflowed content.]

    #lorem(240)

    Mixed scripts are split at the right places: 中文与英文混排 testing mixed CJK
    and Latin reflow.
    假文假文假文假文假文假文假文假文假文假文假文假文假文假文假文假文。

    #lorem(260)
  ]
}

#margin-reflow(
  columns: (count: 1),
  margin: (inside: 1cm, outside: 2.5cm),
)[#reflow-body]
```



Rendered output — page 1 (left) and page 2 (right). Starting from symmetric margins and switching to asymmetric ones.

10 Footnotes and a figure

```
// Example: complex content — footnotes at the start and in the middle, and an
// image, all inside the reflowed block.
//
// Compile to page images with:
//   typst compile -f png --ppi 144 examples/footnotes-figure.typ "doc/images/footnotes-
figure-{p}.png"
//
// The prose switches to Chinese when compiled with `--input lang=zh`, which is
// how `doc/manual-zh.typ` renders its figures.

#import "../lib.typ": margin-reflow
// Removes the spaces that source line breaks would otherwise insert between
// CJK characters.
#import "@preview/cjk-unbreak:0.2.3": remove-cjk-break-space
#import "assets/guides.typ": guides

#let lang = sys.inputs.at("lang", default: "en")

#set page(
  paper: "a5",
  margin: (inside: 1cm, outside: 2.5cm),
  numbering: "1",
  background: guides(),
)
#set text(font: ("New Computer Modern", "SimSun"), size: 9pt, lang: lang)
#set par(first-line-indent: (amount: 2em, all: true), justify: true, leading: 0.6em)
```

```
#set figure(gap: 0.5em)

// Body text at the normal asymmetric margins.
#let preamble = if lang == "zh" { [中文与英文混排的正文示例。] } else { [#lorem(40)] }

#preamble

#v(1em)

#let reflow-body = if lang == "zh" { remove-cjk-break-space[

    #footnote[位于重排内容最开头的脚注。]
    重排内容一开始就带有一条脚注。

    #lorem(220)

    #figure(image("assets/plot.svg", width: 75%), caption: [重排内容中的一张图片。])

    #footnote[位于重排内容中部的脚注。]
    下一段带有一条脚注，其后继续排布文字。

    中文与英文混排 testing mixed CJK and Latin reflow in two columns。
    假文假文假文假文假文假文假文假文假文假文假文假文假文假文假文假文。

    #lorem(240)
] } else {
  [
    #footnote[A footnote at the very start of the reflowed content.]
    The reflowed content starts with a footnote of its own.

    #lorem(220)

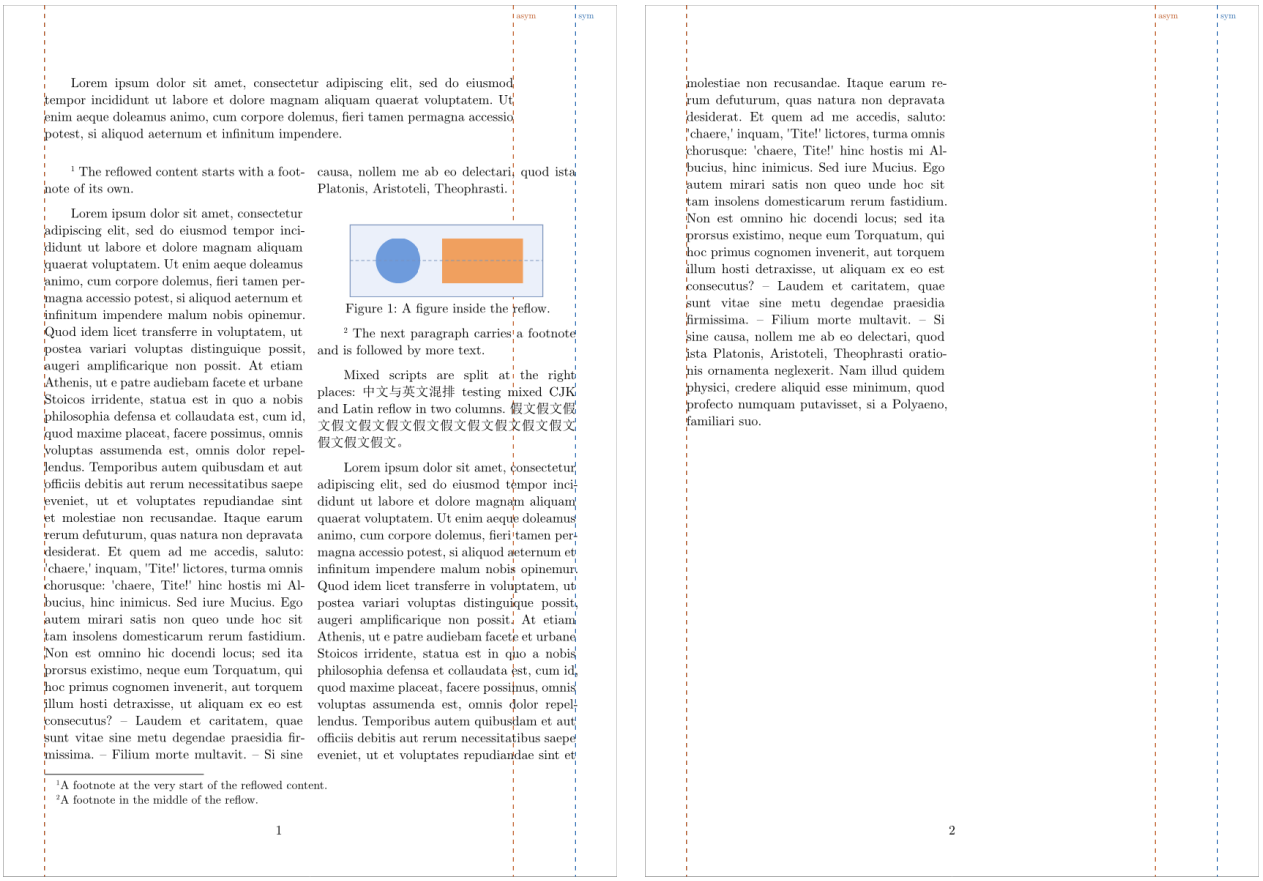
    #figure(image("assets/plot.svg", width: 75%), caption: [A figure inside the
reflow.])

    #footnote[A footnote in the middle of the reflow.] The next paragraph carries
    a footnote and is followed by more text.

    Mixed scripts are split at the right places: 中文与英文混排 testing mixed CJK
    and Latin reflow in two columns.
    假文假文假文假文假文假文假文假文假文假文假文假文假文假文假文假文。

    #lorem(240)
  ]
}

#margin-reflow(
  columns: (count: 2, gutter: 10pt),
  margin: "symmetric",
  footnote-style: (size: 8pt),
)[#reflow-body]
```



Rendered output — page 1 (left) and page 2 (right). A footnote at the very start, a figure in the middle, and a second footnote, all inside the reflowed block.

11 Limitations

11.1 The flow functions cannot be nested

A `margin-reflow` call measures and rewrites the content it is given, and it decides the page geometry from the page state at the position where it appears. A second call **inside** that content cannot be seen by the outer call: the inner call would measure a page whose geometry the outer call has already replaced, and the two margin switches would compete. Keep the calls sequential — call `margin-reflow` again after the first call has ended.

11.2 State updates are not supported inside the reflowed content

The content is laid out several times while the package searches for the amount that fits (and, with `set-page-margin: false`, once per page). Because of that:

- `state` updates inside the reflowed content run more than once and must not be relied upon. A `state` that is **read** elsewhere should be maintained with a metadata-based mechanism instead, not by calling `state.update()` inside the reflow;
- `counter` access that triggers layout-time effects (`counter(footnote)`, `counter(page)`) is handled by the package itself and is fine;
- a manual counter update, such as `#counter("my-counter").update(...)` or `.step()`, is **not** affected by the repeated layout when it is written as plain document content: Typst's counters are resolved after layout, so the final value follows document order.

In short: do not build a `state` machine inside the reflowed content. Do the state handling outside and pass the result in.

11.3 Footnotes are rendered by the package

Footnotes inside the reflow are collected and laid out manually. Typst's own `footnote.entry` rules therefore do not apply to them; use the `footnote-style` parameter for the entry body and `#set footnote.entry(...)` for the spacing. This also means that a footnote that should float to the real bottom of the page (for example with `figure(placement: bottom)` inside the reflow) is placed by the package and not by the page.

11.4 Page setup

- The page width and height must be concrete lengths.
- `set-page-margin: false` requires `count: 1`; with two or more columns the combination is rejected, because a hand-split page cannot switch to a multi-column layout partway through.
- The package builds its own fixed-height `grid`/`block` structure and changes `page.margin`, so layout packages that also control pagination, columns or page geometry may conflict. Packages that only **decorate** content (colours, boxes, emphasis) generally work.
- `margin: auto` inherits the current margin, which on an asymmetric page means the reflowed columns span the asymmetric **width** while being positioned for a symmetric layout. Pass `margin: "symmetric"` (or an explicit dictionary) for a two-column reflow; `auto` is intended for reflowing inside the margins you already have.

12 Compatibility and CJK

12.1 Content-processing packages

The function splits and rebuilds content, so packages that **transform content** should be applied to the content **before** it is passed in. For example, with `cjk-unbreak`:

```
#import "@preview/cjk-unbreak:0.2.3": remove-cjk-break-space

#margin-reflow(
  columns: (count: 2, gutter: 10pt),
  margin: "symmetric",
)[
  #remove-cjk-break-space[
    中文与英文混排 content ...
  ]
]
```

12.2 CJK and mixed scripts

The splitter is aware of Han characters, CJK punctuation and spaces, so it handles Chinese, Japanese and Korean text as well as mixed CJK/Latin reflow. For best results, set a CJK-capable font and enable justification and first-line indentation:

```
#set text(font: ("New Computer Modern", "SimSun"), lang: "zh")
#set par(first-line-indent: (amount: 2em, all: true), justify: true)
```

A line break in the **source** between two CJK characters is rendered as a space. `cjk-unbreak` removes those spaces when the two sides are plain text; it cannot do so across a styled element such as `strong` or a footnote. Keep the line breaks of a CJK paragraph out of the places where a space must not appear.

12.3 Margin-note packages (e.g. marginalia)

Using this package together with a margin-note package such as `marginalia` is a common case, but because it manages the page and changes `page.margin` mid-page, a few precautions are needed:

- Do **not** wrap the output of `margin-reflow` in `marginalia.wideblock`. A wideblock computes its width from the marginalia configuration rather than from the current page margins, and a multi-page wideblock does not work with alternating margins.
- Do **not** use `marginalia.wideblock` or `marginalia.header` for headers and footers. Instead, align the header/footer directly from `page.margin` (a broader alignment that follows the margins this package sets). A minimal example adapted from the book template:

```
#set page(header: context {
  let m = page.margin
  let pad-fn = if "inside" in m {
    if calc.even(here().page()) {
      pad.with(left: m.inside - m.outside, rest: 0pt)
    } else {
      pad.with(right: m.inside - m.outside, rest: 0pt)
    }
  } else {
    it => it
  }
  pad-fn(align(end + horizon)[#counter(page).display()])
})
```

- An **empty** `marginalia.wideblock` may still be used to reserve the widened area so that margin notes do not overflow into the body text.

13 Inspiration

The idea of reflowing content mid-page comes from [meander.typ](https://github.com/Vanille-N/meander.typ). This package focuses on the asymmetric ⇄ symmetric reflow direction and, in particular, on the multi-page asymmetric case that could previously fail to [converge](https://github.com/Vanille-N/meander.typ/issues/1#issuecomment-3306100761). It also supports mixed CJK and Latin reflow.

The name **riffle** stays with the river metaphor: a riffle is the short stretch of a stream where the current changes character — here, the stretch of a page where the layout changes course.

14 License

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